

internal energy can be different because total mass of the gases may not be equal.

13 Ans: B

Using $pV = NkT$,

$$p / kT = N / V$$

$$pm / kT = Nm / V = \rho, \text{ where } m = \text{mass of each molecule}$$

Hence,

$$pm / k(273.15 + 20) = 1.20$$

$$pm / k(273.15 + 27) = \rho$$

Solving,

$$\rho = 1.17 \text{ kg m}^{-3}$$

14 Ans: D